fn sample_bernoulli_exp1

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Proves soundness of `fn sample_bernoulli_exp1` in `mod.rs` at commit 0be3ab3e6 (outdated[1]). `fn sample_bernoulli_exp1` returns a sample from the $Bernoulli(exp(-x))$ distribution for some rational argument in $[0, 1]$. This proof is adapted from subsection 5.1 of [CKS20].

1 Hoare Triple

Preconditions

Compiler-verified

- Argument `x` is of type `RBig`, a rational bignum

Caller-verified

$x \in [0, 1]$

Pseudocode

```
def sample_bernoulli_exp1(x) -> bool:
    k = 1
    while True:
        if sample_bernoulli_rational(x / k): #
            k += 1
        else:
            return is_odd(k) #
```

Postcondition

Theorem 1.1. For any setting of the input parameter `x` such that the given preconditions hold, `sample_bernoulli_exp1` either returns `Err(e)` due to a lack of system entropy, or `Ok(out)`, where `out` is distributed as $Bernoulli(exp(-x))$.

Assume the preconditions are met.

Lemma 1.2. `sample_bernoulli_exp1` only returns `Err(e)` when there is a lack of system entropy.

Proof. In all usages of `sample_bernoulli_rational`, the argument passed satisfies its definition preconditions, by the preconditions on `x` and function logic. Thus, by its postcondition, `sample_bernoulli_rational` only returns an error when there is a lack of system entropy. The only source of errors in `sample_bernoulli_exp1` is from the invocation of `sample_bernoulli_rational`. Therefore `sample_bernoulli_exp1` only returns `Err(e)` when there is a lack of system entropy. □

[1] See new changes with `git diff 0be3ab3e6..481986c20 rust/src/traits/samplers/cks20/mod.rs`

Lemma 1.3. Let K^* denote the final value of `k` on line 7. Then $P[K^* > n] = \frac{x^n}{n!}$ for any integer $n \geq 0$ [CKS20].

Proof. For $k > 0$, let a_k denote the kth outcome of `sample_bernoulli_rational` on line 4. By the definition of `sample_bernoulli_rational`, under the established conditions and preconditions, each A_k is distributed as $Bernoulli(x/k)$.

If $n \geq 1$, we have:

$$\begin{aligned}
P[K^* > n] &= P[A_1 = A_2 = ... = A_n = \top] && \text{since } K^* > n \iff \forall k \leq n, a_k = \top \\
&= \prod_{k=1}^{n} P[A_k = \top] && \text{all } A_k \text{ are independent} \\
&= \prod_{k=1}^{n} \frac{x}{k} && \text{since } A_k \sim Bernoulli(x/k) \\
&= \frac{x^n}{n!}
\end{aligned}$$

If $n = 0$, we also have $P[K^* > 0] = 1 = \frac{x^0}{0!}$. □

Lemma 1.4. $is_odd(K^*) \sim Bernoulli(exp(-x))$ [CKS20].

Proof.

$$\begin{aligned}
P[K^* \text{ odd}] &= \sum_{k=0}^{\infty} P[K^* = 2k+1] \\
&= \sum_{k=0}^{\infty} (P[K^* > 2k] - P[K^* > 2k+1]) \\
&= \sum_{k=0}^{\infty} \left(\frac{x^{2k}}{(2k)!} - \frac{x^{2k+1}}{(2k+1)!} \right) && \text{by 1.3} \\
&= exp(-x)
\end{aligned}$$

□

Proof. Since `k` is distributed according to K^*, then by 1.4, `out` is distributed as $Bernoulli(exp(-x))$. Together with 1.2, Theorem 1.1 holds. □

References

[CKS20] Clément L. Canonne, Gautam Kamath, and Thomas Steinke. The discrete gaussian for differential privacy. *CoRR*, abs/2004.00010, 2020.